

Ariadna Albors Zumel

PHD STUDENT · DEPARTMENT OF SOCIOLOGY AND SOCIAL RESEARCH

University of Trento, Italy

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Education

PhD in Sociology and Social Research

Trento, Italy

UNIVERSITY OF TRENTO

Nov 2022 - present

- Supervisors: Gian Maria Campedelli, Michele Tizzoni
- Thesis: *Advanced Computational Methods for Crime Research: Forecasting using Deep Learning, Causal Inference, and LLM Threat Detection Bias*

MSc in Financial Mathematics

Dublin, Ireland

DUBLIN CITY UNIVERSITY

Sep 2021 – Sep 2022

- Main Courses followed: Probability & Finance (Advanced), Simulation for Finance, Financial & Actuarial Models, Time Series, and Deep Learning.

MSc in Computing Science: Data Science & Systems Complexity

Groningen, The Netherlands

UNIVERSITY OF GRONINGEN

Sep 2019 – March 2021

- Master's thesis supervisor: Michael Biehl
- Thesis: *Analysis of Non-targeted Metabolomics Data using GMLVQ*
- Main Courses: Machine Learning, Machine Learning Theory, Pattern Recognition, Numerical Mathematics, Neural Networks, and Modeling & Simulation.

BSc in Astronomy (Honours)

Groningen, The Netherlands

UNIVERSITY OF GRONINGEN

Sep 2015 – July 2018

- Thesis: *Morphological classification of galaxies using Convolutional Neural Networks*
- Main Courses: Linear Algebra, Calculus 1 & 2, Mathematical Physics, Mechanics and Relativity, Complex Analysis, Numerical Methods, Quantum Physics 1 & 2, and Advanced Electrodynamics.

Professional Experience

Visiting Scholar

Pittsburgh, USA

CARNEGIE MELLON UNIVERSITY, SOFTWARE AND SOCIETAL SYSTEMS DEPARTMENT

Jan 2025 - June 2025

- Visiting supervisor: Kathleen M. Carley
- Followed the course "Dynamic Network Analysis" taught by Prof. Kathleen M. Carley.

Academic Tutor

Trento, Italy

UNIVERSITY OF TRENTO

Sept 2024 - Dec 2024

- Course: Scientific Programming, from the Master in Data Science.
- Tasks: Helped students with their lab exercises on introduction to Python.

Teaching Assistant

Trento, Italy

UNIVERSITY OF TRENTO

March 2024 - June 2024

- Course: Computational Social Science, from the Master in Data Science.
- Tasks: Gave lectures on DAGs, Time Series, and Spatial data analysis.

Data Scientist Intern

Barcelona, Spain

AIZON

Feb 2022 - Nov 2022

- Collaborated with the AI team to support data discovery and solve issues from actual business cases. Also, contributed to the improvement of the platform's AI capabilities.

Online Exam Task Force Assistant

Groningen, The Netherlands

UNIVERSITY OF GRONINGEN

Sep 2020 - July 2021

- Supported instructors in uploading and proctoring their exams online during the COVID-19 pandemic.

Teaching Assistant

UNIVERSITY OF GRONINGEN

Groningen, The Netherlands

Sep 2018 - July 2021

- Conducted the lab exercises sessions for diverse courses from the bachelor degrees of Physics, Astronomy, Mathematics, and Computing Science.

Research Interests

Spatiotemporal crime patterns, Crime forecasting, Human mobility, Deep learning, Machine Learning, LLMs, Crime prevention, Causal inference, Algorithmic bias

Publications

PUBLISHED OR ACCEPTED

Albors Zumel, A., Tizzoni, M. and Campedelli, G.M. Deep Learning for Crime Forecasting: The Role of Mobility at Fine-grained Spatiotemporal Scales. *J Quant Criminol* (2025). <https://doi.org/10.1007/s10940-025-09629-3>

Campedelli, G. M., Jelveh, Z., Chalfin, A., Semenza, D., Piza, E., **Albors Zumel, A.**, Lepri, B., and Sharkey, P. *Nature* 655, 315–326 (2026). <https://doi.org/10.1038/s41586-026-10622-4>

UNDER REVIEW

Albors Zumel, A., Tizzoni, M., Hernández, W., and Campedelli, G.M. Ambient Population and Crime: The Fragility of Causal Links in Urban Environments. Under review at *Justice Quarterly*.

IN PREPARATION

Albors Zumel, A., Tizzoni, M., and Campedelli, G.M. Unequal Suspicion: A Sequential Factorial Vignette Study of LLM Bias in Threat Detection.

Honors & Awards

- Science and Health Dean's Honors List 2023**, Dublin City University
- 2023 Acknowledgment of high academic performance in achieving a first class honors award, as well as being in the top percentage of students in the cohort.
- Honours College student - Einstein cohort**, University of Groningen
- 2018 The top 10% of the 1st-year students of each cohort are invited to follow extracurricular classes representing 45 ECTS.
- Propedeuse BSc Astronomy Cum Laude**, University of Groningen
- 2016 Obtained Cum Laude distinction in the first year of my bachelor's degree.

Presentations & Posters

Sept 2026. *Ambient Population and Crime: The Fragility of Causal Links in Urban Environments*. Oral presentation at: Annual meeting of the European Society of Criminology, Warsaw, Poland.

May 2026. *Ambient Population and Crime: The Fragility of Causal Links in Urban Environments*. Oral presentation at: Italian Conference of Computational Social Science, Torino, Italy.

September 2025. *Deep Learning for Crime Forecasting: The Role of Mobility at Fine-grained Spatiotemporal Scales*. Oral presentation at: Annual meeting of the European Society of Criminology, Athens, Greece.

November 2024. *Mobility Data and Deep Learning for Advancing Spatiotemporal Crime Forecasting in Microgeographic Units*. Poster at: Joint Doctoral Conference, Trento, Italy.

July 2024. *Mobility Data and Deep Learning for Advancing Spatiotemporal Crime Forecasting in Microgeographic Units*. Oral presentation at: International Conference on Computational Social Science, Philadelphia, USA.

Teaching Experience _____

Fall 2024 **Scientific Programming**, Tutor
Spring 2024 **Computational Social Science**, Teaching Assistant
Fall 2020 **Computer Architecture**, Teaching Assistant
Spring 2020 **Introduction to Python**, Teaching Assistant
Fall 2019 **Linear Algebra**, Teaching Assistant
Fall 2019 **Advanced Algorithms and Data Structures**, Teaching Assistant
Spring 2019 **Mathematical Physics**, Teaching Assistant
Fall 2018 **Calculus 2**, Teaching Assistant
Fall 2018 **Calculus 1**, Teaching Assistant

Reviewed for _____

Journal of Quantitative Criminology, EPJ Data Science, and Frontiers in Big Data

Language proficiency _____

Catalan (native language), Spanish (bilingual proficiency), English (fluent in speaking, reading, and writing), Italian (fluent in reading and writing, good in speaking), Dutch (basic comprehension)

Programming languages _____

Python, R, SQL, HTML, CSS